

When “Shortcut-Breaking” Splits Leak: A Source-Trust-Isolating Benchmark for Fact Verification

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Abstract—Fact verification (FV) models are known to exploit shortcuts such as claim-only lexical bias, and a natural remedy is to weight evidence by the trustworthiness of its source. We study how such source-trust reasoning is measured, and report three problems in a representative source-trust FV system. First, its synthetic “shortcut-breaking” split, intended to require source trust, is solved at 0.996 accuracy by a trust-blind model, because low-trust evidence is drawn from a different textual register (writing style) than high-trust evidence: a single template field predicts the verdict at 0.99, so text style alone is sufficient. Second, its reported AVeriTeC accuracy of 0.665 equals the majority-class prior (macro-F1 0.265). Third, its simulation split reaches accuracy 1.000 through closed-world triviality. We then construct a trust-isolating benchmark that admits no detectable non-trust path to the label, by holding the evidence text byte-identical across trust tiers, holding the source category constant, and holding out trust values at test time. The result is a sharp inversion: a trust-blind model that won on the leaky split collapses to chance (graph-network macro-F1 0.444), while a trust-aware model wins (macro-F1 0.772 with the original graph network, and 1.000 with a controlled linear probe, an isolation check rather than a capability claim). We release the benchmark and its generator at <https://github.com/AISWARYA-NANDAKUMAR/epistemic-factkg-benchmark>.

Index Terms—fact verification, source credibility, shortcut learning, dataset leakage, benchmark design, graph neural networks

I. INTRODUCTION

Automated fact verification predicts whether a claim is supported, refuted, or unverifiable given retrieved evidence. A persistent failure mode is shortcut learning: models reach high accuracy by exploiting surface regularities rather than reasoning over evidence. The best known case is claim-only bias in FEVER, where a classifier that never reads the evidence nearly matches an evidence-aware model [1]. A natural and well-motivated direction is to make verification *source-aware*: identical text from a peer-reviewed article and from an anonymous post should not be treated identically, so evidence should be weighted by the trustworthiness of its source [2], [3].

This paper is about how that idea is *measured*. To claim that a model uses source trust, one needs an evaluation that only a trust-using model can solve. The intuitive construction pairs the same claim with a high-trust and a low-trust source and labels the low-trust version not-enough-evidence. We show that a careful instance of this construction silently leaks the

verdict, that the leakage makes a trust-blind model appear to succeed, and that a second common reporting choice, accuracy on a class-imbalanced benchmark, hides a majority-class predictor. We then show how to build an evaluation that does isolate source trust, and what changes when it is used.

Our study object is a representative source-trust FV system that computes a per-evidence epistemic confidence from source trust, an evidence-type weight, and an inference strength, and aggregates these into a verdict through a heterogeneous graph neural network with a symbolic and a neural decision layer. We audit its evaluation, find it does not measure what it claims, and rebuild the measurement. Fig. 1 summarizes the corrected benchmark.

Contributions.

- 1) **A leakage diagnosis.** We show that a synthetic “shortcut-breaking” trust split leaks the verdict through textual register (a template field predicts it at 0.99 and a trust-blind model scores 0.996), that reporting AVeriTeC accuracy of 0.665 masks the majority-class prior (macro-F1 0.265), and that the simulation split is closed-world trivial (Section IV).
- 2) **A trust-isolating benchmark.** We construct a split that admits no detectable non-trust path to the label, and report the lesson that text isolation alone is insufficient: a first version still leaked at macro-F1 0.82 through the source-category proxy until the category was neutralized (Section V).
- 3) **The inversion.** On the corrected benchmark a trust-blind model is at chance and a trust-aware model wins, the opposite of the leaky split, confirmed with both a controlled probe and the original graph network (Section VI).

We release the benchmark generator and data so the construction can be inspected and extended.

II. BACKGROUND AND RELATED WORK

Fact-verification benchmarks. FEVER introduced large-scale claim verification over Wikipedia with three verdict classes [4]. AVeriTeC moved to real-world claims with web evidence and four verdict classes, and is heavily skewed toward the refuted class [5]. AVeriTeC distinguishes plain verdict accuracy from an evidence-gated score; we use it only as a verdict-accuracy testbed and report macro-F1 against the

majority baseline, because the class skew makes accuracy alone uninformative.

Shortcuts and robustness in FV. Claim-only bias was documented for FEVER together with a symmetric evaluation set [1]. VitaminC built contrastive evidence pairs that force a model to read the evidence rather than the claim [6]. These works target lexical and evidence-content shortcuts. The shortcut we study is different: a model can recover the verdict from the *register* of the evidence text, which is correlated with source trust by construction. Our corrected benchmark is the source-trust analogue of a contrastive split.

Source-credibility verification. DeClarE weighted evidence with a learned source-trustworthiness representation for claim credibility [2]. Source factuality has been predicted at the outlet level from multiple signals [3]. These supply or use a trust signal but do not provide a measurement that isolates it; our contribution is the isolating evaluation.

Graph and symbolic FV. Evidence aggregation over graphs is a standard FV backbone, from GEAR [7] and the kernel graph attention network [8] to stance-aware aggregation [9] and heterogeneous-graph reasoning over text and tables [10]. Rule-based aggregation over per-evidence entailment has also been used for grounding checks [11], and logic-regularized models give interpretable, auditable verdicts [12]. The system we audit sits in this family; our focus is its evaluation, not its architecture. Epistemology-grounded reasoning in language models is nascent and, to date, targets general deduction rather than fact verification [18].

III. THE SYSTEM UNDER STUDY

The audited system is our own earlier implementation, Epistemic FactKG (system code at <https://github.com/karkid/epistemic-factkg>); its leaky-split, AVeriTeC, and simulation numbers (the “before” column of Table III and the upper rows of Table I) are taken from that implementation’s evaluation logs, with the corrected benchmark released at the repository above. The system assigns each evidence item an epistemic confidence

$$EC_i = 1 - (1 - ST_i)^{EW_i \cdot IS_i}, \quad (1)$$

where $ST_i \in [0, 1]$ is a registry source-trust score, EW_i is an evidence-type weight, and IS_i is an inference strength. Per-claim evidence is encoded by a heterogeneous graph network (graph attention over claim and evidence nodes) with stance and inference-strength heads, and the verdict is produced either by a symbolic threshold on the aggregated EC or by a learned verdict head. For this study the relevant property of Eq. (1) is monotonicity in ST: with evidence type and inference strength fixed, raising source trust raises confidence. A model that ignores ST has no access to this signal. The evidence-type weight EW is an orthogonal axis that our source-trust benchmark deliberately holds fixed; we return to it in Section VII.

TABLE I
DIAGNOSIS OF THE ORIGINAL EVALUATION. EACH ROW REPORTS A NUMBER THAT UNDERCUTS A SOURCE-TRUST CLAIM. LOOKUP VALUES ARE THE FRACTION OF VERDICTS RECOVERED FROM A SINGLE NON-TRUST FEATURE.

Finding	Measure	Value
Synthetic: template field	verdict lookup	0.99
Synthetic: source id	verdict lookup	0.82
Synthetic: trust-blind model	accuracy ↑	0.996
AVeriTeC: majority baseline	accuracy / macro-F1	0.660 / 0.265
AVeriTeC: text classifier	accuracy / macro-F1	0.656 / 0.354
Simulation split	accuracy ↑	1.000

IV. DIAGNOSIS: THE ORIGINAL EVALUATION DOES NOT MEASURE SOURCE TRUST

We evaluate three claims in the original setup and find each undermined by its own data. Table I collects the numbers.

The synthetic split leaks through text register. The synthetic “shortcut-breaking” split pairs claims with high- and low-trust sources and labels low-trust supporting text as not-enough-evidence, so that, in principle, only source trust can separate the pair. In practice the generator draws low-trust evidence from a rumour register (for example, “An anonymous post claimed that...”) and high-trust evidence from an official register (“published findings confirming...”). Because the register is determined by the trust tier and the trust tier determines the verdict, the text register determines the verdict. A single-feature lookup confirms this: a per-record template field predicts the verdict at 0.99, the source identifier at 0.82, and the raw evidence text and its stance at 0.99 and 0.54 respectively (Table I, Fig. 2). Consequently a trust-blind model that reads only the text reaches 0.996 accuracy on this split, and the trust-aware model scores lower (0.889), because the leaky split rewards reading the register rather than reasoning about the source.

AVeriTeC accuracy hides the class prior. The AVeriTeC subset is heavily imbalanced. On the three-class development set, always predicting the majority class (refuted) yields accuracy 0.660 and macro-F1 0.265. A standard text classifier collapses onto the majority class, predicting the not-enough-evidence class exactly once out of 462 examples, for accuracy 0.656 and macro-F1 0.354. The reported system accuracy of 0.665 lies on this prior; without macro-F1 it cannot be distinguished from a majority predictor.

The simulation split is closed-world trivial. On the simulation split, every evidence item has the same maximal source trust and evidence-type weight, so Eq. (1) carries no discriminative signal, and the verdict is recoverable by matching a numeric value against simulator ground truth. Accuracy is 1.000, which reflects task triviality rather than epistemic reasoning, and inflates any average that includes it.

V. A TRUST-ISOLATING BENCHMARK

We rebuild the evaluation so that source trust is the only feature that determines the label. Fig. 1 shows the construction.

TABLE II

TRUST-ISOLATION VERIFICATION ON THE CORRECTED BENCHMARK. A NON-LEAKING FEATURE PREDICTS THE VERDICT AT THE CHANCE RATE (0.50 FOR THE HIGH/LOW DECISION). SOURCE TRUST IS THE ONLY PREDICTIVE SIGNAL, AND IT IS HELD OUT BY VALUE AT TEST.

Feature	verdict predictable
evidence text (full)	0.50
evidence text (prefix)	0.50
stance	0.50
source category	0.50 (constant)
source-trust scalar	predictive (held out at test)

Design. For each claim we generate one neutral evidence sentence and attach it, byte-for-byte identical, to a high-trust and a low-trust source. The verdict is derived from the system’s own confidence rule, Eq. (1): with a single testimony item, high trust ($ST \geq 0.85$) yields supported or refuted, and low trust ($ST \leq 0.45$) yields not-enough-evidence. The text is therefore constant within each high/low pair, and only the source-trust scalar flips the label. The benchmark has 2080 records, 1600 for training and 480 held out for test.

Text isolation is necessary but not sufficient. A first version of the benchmark held the text identical across tiers but allowed sources of different categories. It looked clean under single-feature checks, yet a trust-blind model still reached macro-F1 0.82 on the held-out test. The reason is an interaction the single-feature checks missed: the model reads the 6-dimensional source-category feature, category is correlated with trust, and so it transfers to most held-out sources; combined with a “supporting text implies supported” default, it recovers the label without using the trust scalar. We therefore hold the source *category* constant (a single category for all records) and vary only the continuous trust scalar, with held-out trust values at test so that a trust-using model must treat trust as a continuous signal rather than memorize values. This is itself a lesson: isolating a feature requires controlling its proxies, not only the feature.

Verification. The corrected benchmark admits no non-trust path to the label. Every distinct evidence text appears with more than one gold label (747 of 747), and every non-trust feature predicts the verdict at chance (Table II). Fig. 2 contrasts the per-feature leakage of the original and corrected splits.

VI. EXPERIMENTS: THE INVERSION

We evaluate trust-blind and trust-aware models on the corrected benchmark, with the majority and all-not-enough-evidence baselines, and macro-F1 as the primary metric because of the class balance by construction. The text embedding is all-MiniLM-L6-v2 (384 dimensions); the trust-blind baseline and the trust-aware model are the audited system’s own heterogeneous graph attention network, the latter with the epistemic-confidence layer that reads source trust. The benchmark generator is seeded for exact reproduction.

Controlled probe. To isolate feature access from model capacity, we train a logistic-regression probe on the same

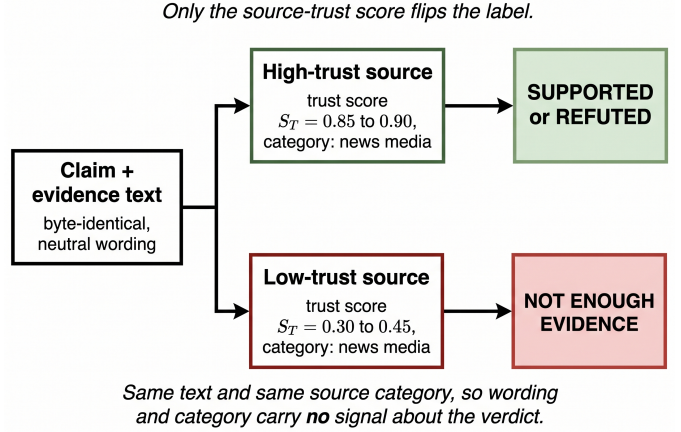


Fig. 1. The trust-isolating benchmark. The same evidence text is attached to a high-trust and a low-trust source of the same category; only the source-trust scalar differs, so text and category carry no label signal and only a trust-using model can separate the pair.

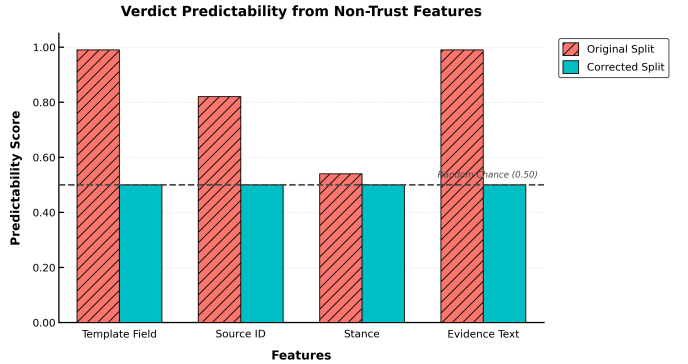


Fig. 2. Verdict recoverable from a single non-trust feature. On the original split the template field, source identifier, evidence text, and stance all leak the verdict (0.99, 0.82, 0.99, 0.54); on the corrected split every non-trust feature sits at chance (0.50).

features the trust-blind graph network sees (a sentence embedding of the evidence text and a source-category indicator), and compare it to the same probe with the source-trust scalar added. On the held-out-trust test set, the trust-blind probe reaches macro-F1 0.447, at chance, while the trust-aware probe reaches 1.000. The perfect score is not a capability claim: it shows that the trust scalar is necessary and sufficient on this benchmark, which is exactly the isolation property we sought.

Original graph network. We then train the system’s own models. The trust-blind baseline (a heterogeneous graph network with a verdict head, with no access to the trust scalar) plateaus at verdict accuracy near 0.50 in training and reaches macro-F1 0.444 on the held-out test, at chance. The trust-aware model (the same backbone with the epistemic-confidence layer, which reads source trust) reaches macro-F1 0.772. The gap of 0.328 macro-F1 is produced solely by access to source trust. The trust-aware model does not reach

TABLE III

THE INVERSION. TRUST-BLIND VERSUS TRUST-AWARE MODELS ON THE LEAKY SPLIT (ACCURACY) AND THE CORRECTED SPLIT (MACRO-F1). BEST RESULT IN EACH SETTING IN BOLD. TRUST-BLIND WINS ON THE LEAKY SPLIT AND COLLAPSES TO CHANCE ON THE CORRECTED ONE.

Setting	Metric	Trust-blind	Trust-aware
Original split (leaky)	accuracy $\uparrow$	0.996	0.889
Corrected, controlled probe	macro-F1 $\uparrow$	0.447	1.000
Corrected, graph network	macro-F1 $\uparrow$	0.444	0.772

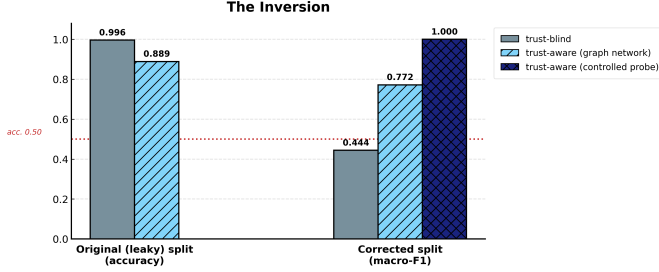


Fig. 3. The inversion. On the original leaky split a trust-blind model wins on accuracy; on the corrected split it collapses to chance (macro-F1) while a trust-aware model wins, with both the controlled probe and the original graph network.

the probe’s perfect score, because its threshold-and-fallback decision generalizes imperfectly to unseen trust values; we report this rather than tune it away.

The inversion. Table III and Fig. 3 place the results beside the original leaky split. On the leaky split the trust-blind model wins (0.996 versus 0.889 accuracy), because the split rewards reading the register. On the corrected split the trust-blind model collapses to chance and the trust-aware model wins. The trust signal helps only once the evaluation stops leaking. This inversion is the central result: it shows both that the original evaluation measured the wrong thing and that, measured correctly, source trust does carry the claimed effect.

VII. LIMITATIONS AND FUTURE WORK

The corrected benchmark is a controlled instrument. It is single-evidence, single evidence-type, fictional-claim, and single-category by design, precisely so that source trust is the only signal; it therefore measures trust sensitivity rather than open-web realism, and the realism axis is better carried by benchmarks such as AVeriTeC reported with macro-F1 and a majority baseline. The trust-aware probe score of 1.000 reflects a label that is a deterministic function of source trust on this split; the scientific content is the contrast with the trust-blind 0.447, not the absolute value. Our experiments use a single training run; multiple seeds with significance tests should be added for a camera-ready comparison. Finally, the evidence-type weight EW in Eq. (1), the axis that the system grounds in an evidence-type taxonomy, is held constant here and is therefore untested; an analogous evidence-type-isolating benchmark, holding source trust fixed while varying evidence

type, is the natural next step and would determine whether that taxonomy is load-bearing.

VIII. CONCLUSION

Source-trust reasoning in fact verification must be measured on evaluations that demonstrably require source trust. We showed that a natural “shortcut-breaking” construction leaks the verdict through textual register, that accuracy on a skewed benchmark hides the majority-class prior, and that controlling a feature requires controlling its proxies as well. We released a trust-isolating benchmark and reported an inversion: a trust-blind model wins on the leaky split and is at chance on the corrected one, while a trust-aware model wins only when the evaluation no longer leaks. The same method should be applied to the orthogonal evidence-type axis before the broader epistemic framing can be claimed.

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